Finite-Source Cosmology from Observer-Patch Holography

Source, Transfer, and Likelihood Boundaries

B. Müller

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Author affiliation: Bernhard Mueller, Pragma Research Inc.

Abstract

Observer-patch holography (OPH) supplies a conditional source-screen theorem. A primitive collar law fixes the angular amplitude, an edge-center generator fixes the tilt, and the radial packet proves the limits of one-shell inversion together with physical dilation and tomography routes to uniqueness. A physical cosmology prediction also requires one finite source instantiation, Boltzmann transfer, and a complete likelihood. Spectra and sky comparisons without that package are diagnostics.

1 Claim Boundary

A physical cosmological prediction requires one construction that maps finite OPH records to a covariant source, propagates that source through a declared background, and evaluates the resulting observables with a complete likelihood. No such construction is supplied. Simulator outputs may test numerical interfaces and expose incompatible assumptions. They carry no cosmological verdict.

The same source premises govern the dark-sector, cosmological-vacuum, and compact OPH constructions.

2 Finite-Source Interface

Let X_r denote a finite observer-screen state at refinement level r. A cosmological source map must emit

$$\mathsf{S}_r(X_r) = \left(g_{\mu\nu}, T_{\mu\nu}, \mathcal{I}, \mathcal{B}\right)_r, \tag{1}$$

where $T_{\mu\nu}$ is conserved, $\mathcal{I}$ contains gauge-invariant initial data, and $\mathcal{B}$ states the background and boundary conditions. The map must be defined without using the observed sky quantities that its outputs are compared with.

The complete one-shell covariance determines the three-dimensional correlation only for chord lengths $0 \leq s \leq 2R_\star$. The resulting radial kernel is infinite-dimensional and persists under positivity. A source-derived lift therefore requires either a scale-natural embedding that transports refinement to physical log-wavenumber dilation or complete radial cross-covariances that support spherical-Hankel tomography. The physical source map must also specify the radial measure, null-space treatment, clock, normalization, and non-fitting forward residual. No finite source construction satisfying these conditions is supplied here. Its source-dependency graph must be finite and acyclic. This graph is a scientific prerequisite graph among stress, clock, scale, radial, transfer, and likelihood objects. It is distinct from the semantic read-from order of the consensus layer. If its nodes are instantiated by semantic commits, a declared mapping and complete certified read supports can make the generated informational order verify their execution dependencies. The provenance theorem alone derives neither these scientific nodes nor their prerequisite edges. The exact same-carrier tensor theorem is narrower than the cosmology graph. Inside a supplied $3+1$ tensor interface, nine supplied balances on fixed algebraic inverse images of a coordinate tomography frame imply all-null balance and an Einstein-form identity. That distinguished nine-set and the tensors are not selected by the informational order, the set is not invariant under all Lorentz or SO(3) transformations, and the universal-source equality carried by the bundle is not used in the tensor equation. The theorem constructs no premise inhabitant, event dimension, physical curvature, transfer function, or likelihood object. The imposed FCC/O_h causet fixture likewise supplies only exact nested provenance, checked cubic carrier-ball matches for $0 \leq r \leq 20$, checked quartic fine-carrier matches for $1 \leq n \leq 10$, and a bounded ordering fraction whose near-four dimension label is its inversion, not independent evidence. The count matches are bounded, not all-radius or all-depth theorems. Its profile trend reverses in an additional out-of-family depth-24 extrapolation and its matched-profile comparison fails. It selects no physical event map, primordial source, radial lift, transfer system, or cosmological likelihood, and therefore changes none of the unsupplied interfaces below.

2.1 Conditional source spectrum

On a source-stress-complete uniform-density release cut, the screen field is

$$q_r = \Pi_{\geq 2,r}\left[\frac{1}{3}\log\frac{J_{X,r}}{\bar{J}_{X,r}}\right]. \tag{2}$$

The primitive collar ledger selects a relative-MaxEnt release law before the screen energy is evaluated. It excludes spectra, residuals, likelihoods, fitted amplitudes, and measurement-calibrated proxies. The release energy then fixes

$$A_{q,r} = \frac{1}{d_r}\mathbb{E}_{\nu_r^{\mathrm{rel}}}[q_r^{\mathsf{T}} M_r K_{\theta,r} q_r] = \frac{2E^{\mathrm{src}}_{q,r}}{d_r}. \tag{3}$$

The conformal precision and angular spectrum are

$$K_\theta = \frac{\Gamma(\sqrt{-\Delta_{S^2}+1/4}+3/2+\theta/2)}{\Gamma(\sqrt{-\Delta_{S^2}+1/4}-1/2-\theta/2)}, \tag{4}$$

$$C_\ell^q = A_q\frac{\Gamma(\ell-\theta/2)}{\Gamma(\ell+2+\theta/2)}, \qquad \ell \geq 2. \tag{5}$$

A required infinitesimal reserve-generator receipt must supply a full-collar density $P_\star/24$. A separately proved weighted orientation-reversal coarea identity then supplies its source-facing half, so

$$\theta = \frac{P_\star}{48}, \qquad n_s = 1 - \frac{P_\star}{48}, \qquad \kappa^{\mathrm{edge}}_{\mathrm{rep}} = \frac{P_\star}{48(P_\star-\varphi)}. \tag{6}$$

A finite one-step survival probability has exponent $-\log u_q(\log b)/\log b$ and does not supply this infinitesimal density.

On the source-dilation branch, the finite source embedding must satisfy

$$D_{s,r}U_r = U_r R_{s,r}, \qquad C_{\zeta,r} = U_r C_{\mathrm{src},r} U_r^*. \tag{7}$$

Strong finite-to-continuum convergence, a uniform covariance norm bound, and a vanishing operator residual then give

$$D_s^{-1} C_\zeta D_s = e^{-\theta s} C_\zeta. \tag{8}$$

This identity forces

$$\Delta_\zeta^2(k) = A_\zeta (k/k_\star)^{-\theta}, \qquad A_\zeta = \frac{A_q}{\pi^{3/2} Z_q^2 (k_\star R_\star)^\theta} \frac{\Gamma(3/2+\theta/2)}{\Gamma(1+\theta/2)}. \tag{9}$$

The tomography branch obtains uniqueness from complete radial cross-covariances. A finite prior selects a conditional radial continuation. Every finite branch publishes its singular spectrum, null basis, finite-window error, positivity check, and forward residual. Temperature and polarization spectra belong to the transfer interface.

3 Dark-Sector Interface

The dark-matter paper proposes a repair-charge rotor with occupation n and phase θ. Conditional on that action, a source-free dilute branch has

$$n \propto a^{-3}, \qquad \rho_R \propto a^{-3}, \qquad w_R \simeq 0. \tag{10}$$

Its cubic condensed branch gives the deep radial-acceleration law and the baryonic Tully–Fisher scaling under static spherical premises.

Equation (10) does not determine the abundance, initial perturbations, relativistic stress, gravitational slip, sound speed, nonlinear evolution, or coupling to the visible sector. A cosmological use requires a covariant completion that emits these quantities from the same action and obeys the full energy-momentum ledger. That completion is not supplied here.

4 Primordial Interface

A physical primordial branch must supply all of the following:

1. a finite source ensemble and a clock;
2. a conserved covariant stress tensor;
3. gauge-invariant scalar, vector, and tensor initial data;
4. a screen-to-volume lift with a controlled null space;
5. phase, amplitude, and isocurvature conditions;
6. refinement stability and an error model.

The conditional source theorem selects $n_s = 1 - P_\star/48$ from the infinitesimal edge-center generator receipt. Physical primordial use requires one finite source construction that supplies this receipt together with the release law, stress, clock, freezeout, mode, scale, and radial receipts. Numerical agreement with a measured summary does not supply the missing source map.

5 Transfer Interface

Given a completed source map, the transfer calculation must declare the background species, equations of state, collision terms, recombination model, initial modes, gauge convention, numerical tolerances, and observable projection. Standard Einstein–Boltzmann software can test this interface with imported inputs. Such a run tests the solver path rather than an OPH source theorem.

The minimum transfer outputs are

$$(C_\ell^{TT}, C_\ell^{TE}, C_\ell^{EE}, C_L^{\phi\phi}, P_m(k,z), H(z), D_A(z), f\sigma_8(z)). \tag{11}$$

All outputs must arise from one compatible background and perturbation system.

6 Likelihood Interface

A data comparison must state the datasets, masks, covariances, nuisance parameters, priors, and combination rule. Overlapping samples and shared calibrations must be represented in the covariance model. Visual overlays, diagonal residual sums, and comparisons made after selecting a formula are diagnostics.

The cosmic microwave background (CMB), galaxy, and background comparisons are diagnostic. They define no cosmological falsifiers or forward targets.

7 Theorem scope

Object	Classification
Finite observer-screen carriers	Conditional finite construction
Screen scalar covariance	Conditional theorem
Source-functional screen amplitude and edge-center tilt	Conditional theorem; finite receipt not supplied
Radial-lift mathematics	Conditional theorem packet; one-shell unrestricted inversion is impossible
Physical source-dilation or radial-tomography receipt	Not supplied
Primordial covariant source	Not supplied
Repair-charge cosmological abundance	Not supplied
Relativistic repair stress and slip	Not supplied
Boltzmann initial-state bridge	Not supplied
Joint CMB, baryon acoustic oscillation, lensing, growth, and cluster likelihood	Not supplied
Cosmological falsification target	No target or falsification verdict is defined

8 Conclusion

Finite OPH screen data become cosmology only through a covariant source map, a three-dimensional lift, a conserved stress tensor, compatible transfer equations, and a complete likelihood. These constructions are not supplied here. Existing cosmological calculations are diagnostics and carry no falsification verdict.

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